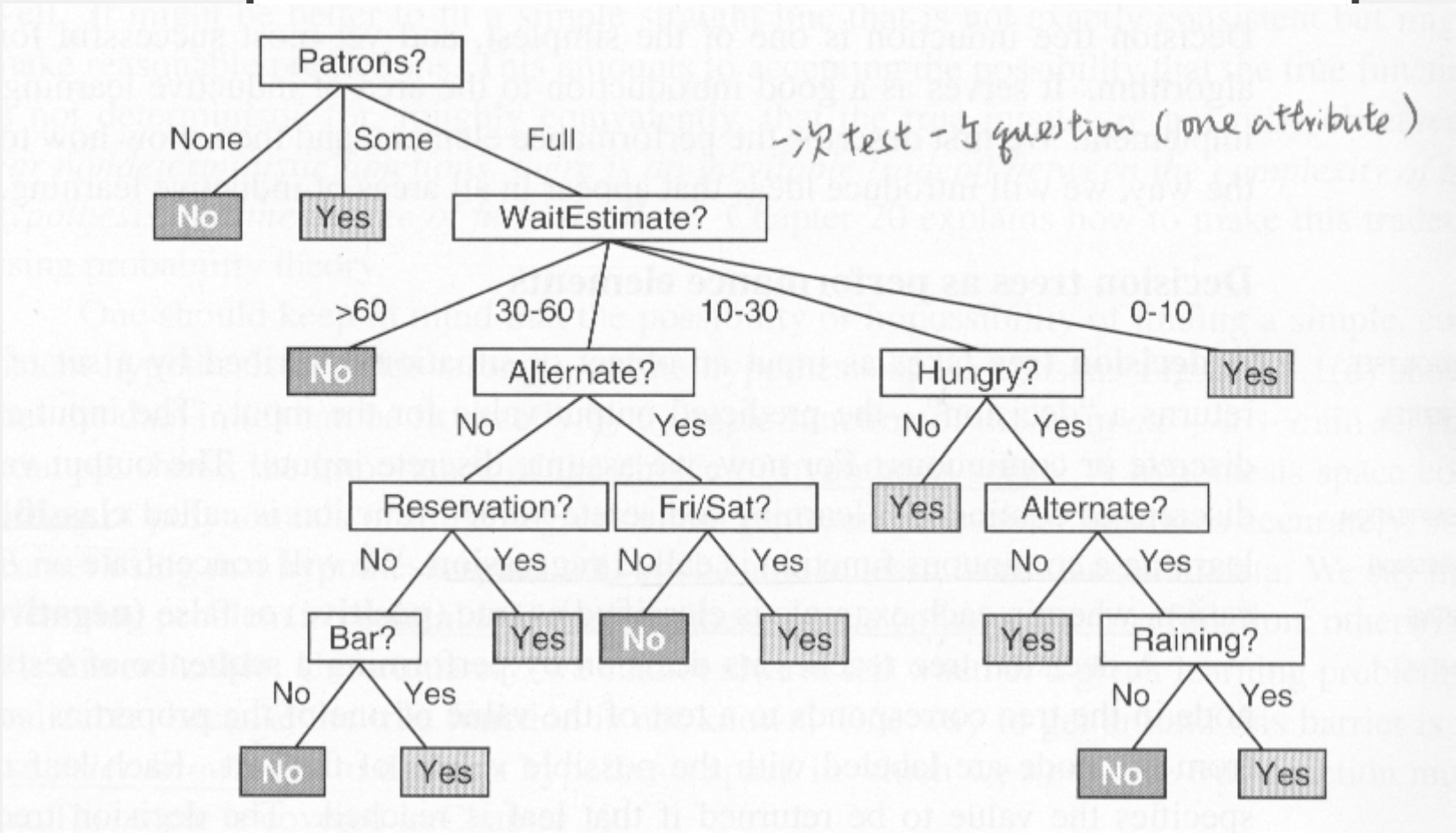
INDUCTIVE LEARNING - CLASSIFICATION AND REGRESSION TREE

DECISION TREE

Cited from AI, A modern approach, Russel Norvig



一次test一个question (one attribute)

DECISION TREE

- Every non-leaf node is attached with a question based on some property
- Every leaf node is assigned a class
  - Yes/No here
- Every branch onto a node is associated with a condition for the question applied at its parent node
- Every instance can be classified to specific class after going through the decision tree
- Is helpful for decision making
- It can be generated manually or automatically

INDUCTIVE METHODS

- What is induction?
  - Find common rules from cases/experiences
- Human is able to use inductive methods
  - Find classification rules for things
  - Construct ontologies for things (classification trees)
- Induction can help to make decision
  - Apply a school: reputation, site, fee, gender, ….
  - Find mates: economy, character, look, shape, …
  - Find jobs: money, load, distance, prospect, …
  - Invest: fund, risk, reward, value, …
  - Buy a house/car or choose restaurants …
- Can computers perform inductive learning?

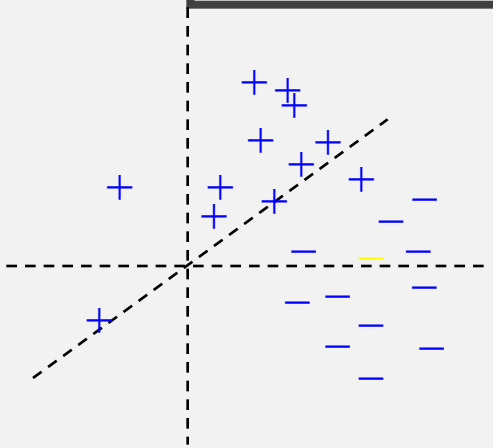
CASES FOR DECISION

Input $\underline{X}$

Output Y

	income	height	weight	look	school	PASS
A	32k	172	66	A	X Univ.	YES
B	35k	166	56	B	C Univ.	NO
C	29k	180	79	B	T Univ.	YES
D	25k	175	65	A	U Univ.	NO
E	40k	169	75	C	Y Univ.	NO
…	…	…	…	…	…	…

VECTOR SPACE MODEL



- Input X (domain): high dimensional vector
  - $\underline{X}$ = [32k, 172, 66, A+, X-Univ]
- Output Y(range): YES(+)/NO(-)
  - Y = [YES]
- Y = f($\underline{X}$) f: classifier function

PRACTICE: VECTOR SPACE MODEL

- Represent a decision issue as vector space model
  - Input X, output Y
  - Question, Property
- Example
  - Apply a school
  - Find a mate
  - Look for a job
  - Perform investment
  - Buy a car
  - Buy a house

	Y	X_1	X_2	X_3	X_4
	RISK	Credit history	Debt	Collateral	Income
E1	**High**	Bad	High	None	<15k
E2	**High**	Unknown	High	None	15k-35k
E3	**Moderate**	Unknown	Low	None	15k-35k
E4	**High**	Unknown	Low	None	<15k
E5	**Low**	Unknown	Low	None	>35k
E6	**Low**	Unknown	Low	Adequate	>35k
E7	**High**	Bad	Low	None	<15k
E8	**Moderate**	Bad	Low	Adequate	>35k
E9	**Low**	Good	Low	None	>35k
E10	**Low**	Good	High	Adequate	>35k
E11	**High**	Good	High	None	<15k
E12	**Moderate**	Good	High	None	15k-35k
E13	**Low**	Good	High	None	>35k
E14	**High**	Bad	High	None	15k-35k

PRACTICE: PARTITION OF DATA

- Please partition the data E1~E14 according to the property X1 (credit history)
- Please partition the data E1~E14 according to property X4 (income)

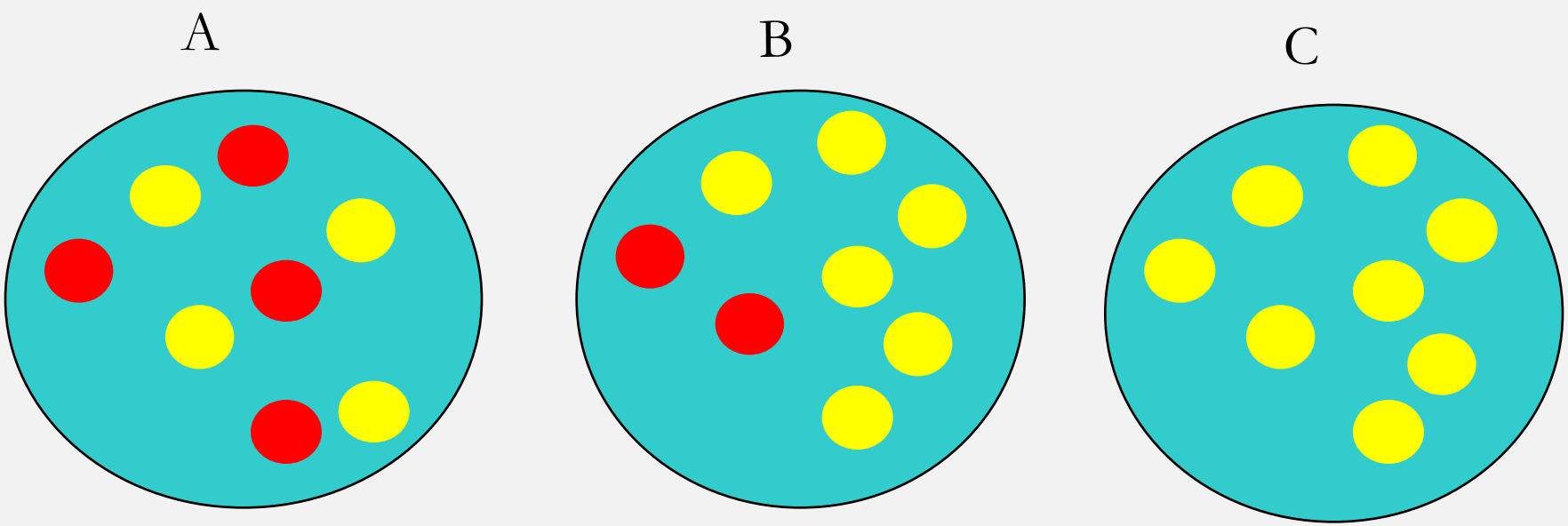
ID3 INDUCTIVE LEARNING

- Generate decision tree according to the training data automatically
- How?
  - Apply a question about some property for a tree node
    - e.g.: "What is the income of the customer?"
  - The training data could be partitioned based on some question
    - e.g. < 15k, 15k-35k, > 35k
  - Which question should be asked first ?

    → according to entropy!

ENTROPY: MEASURE OF UNCERTAINTY

- Degree of chaos for a distribution
- Uncertainty about the observation (picking a ball)



Entropy high ~
Uncertainty high

Entropy low ~
Uncertainty low

ENTROPY

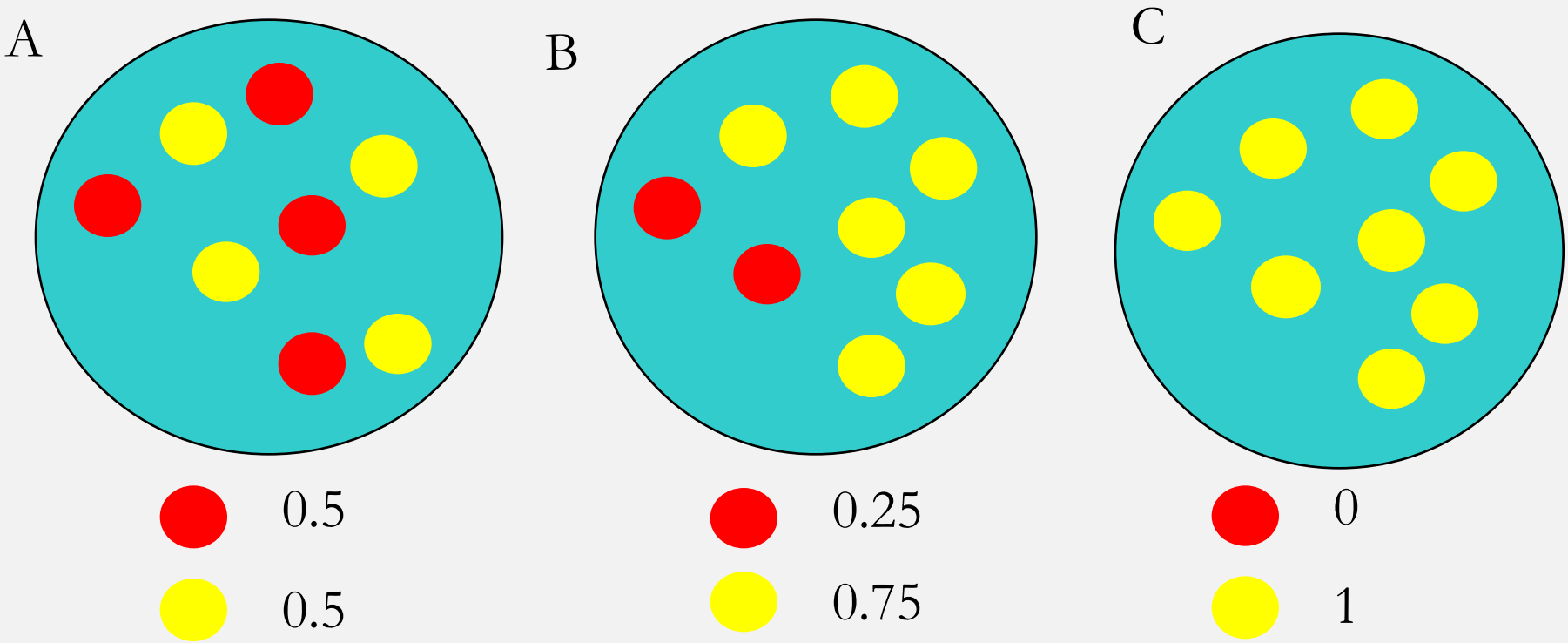
- Definition(Information Theory, Shannon)
  - For distribution $p(m_i)$ of $M = \{ m_1, m_2, \ldots, m_n \}$

    $I(M) \equiv \Sigma_i\, p(m_i)\log_2(1/p(m_i)) = -\Sigma_i\, p(m_i) \cdot \log_2(p(m_i))$
- Measure of average amount of information
  - Information for the outcome m_i: $\log_2(1/p(m_i))$
  - The lower $p(m_i)$ is, the more the information
    - e.g. "finding living creature on Mars"
  - probability=1 → no information (entropy = 0)
    - A MUST has no news-value

      e.g. "the sun rises from the east

ENTROPY: MEASURING DEGREES OF CHAOS / PURIFICATION

- the lower the entropy for a set is, the higher the homogeneity



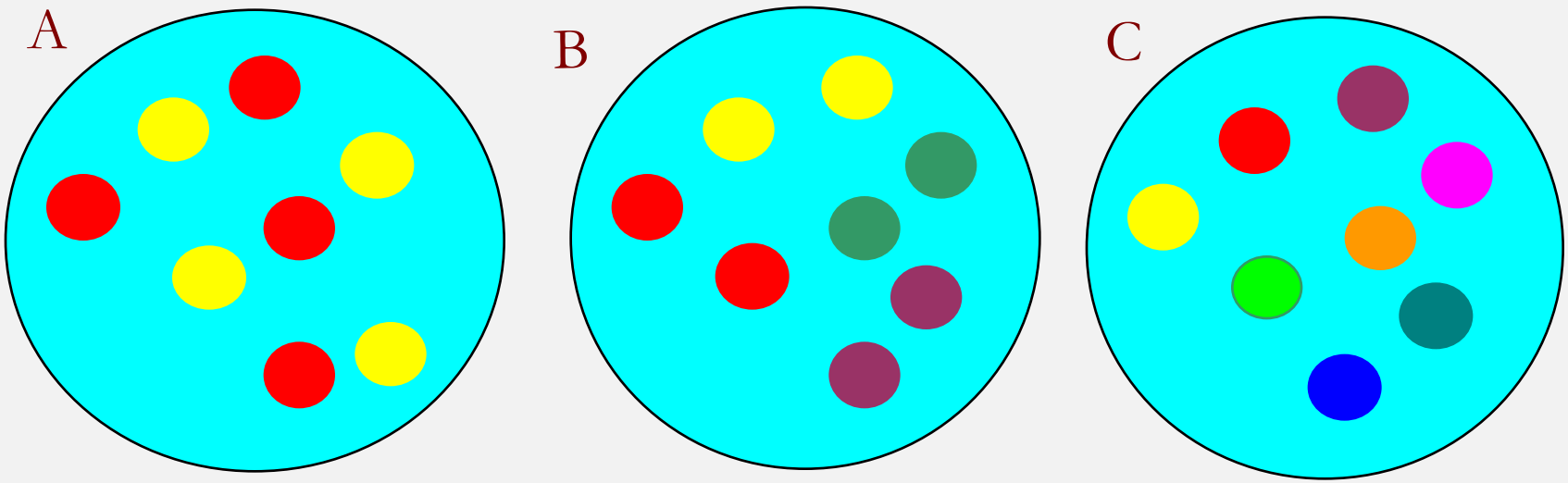
$I(A) = -0.5 \cdot \log_2(0.5) - 0.5 \cdot \log_2(0.5) =$ **1 bit**

$I(B) = -0.75 \cdot \log2(0.75) - 0.25 \cdot \log2(0.25)$ $=$ **0.811 bit**

$I(C) = -1 \cdot \log2(1)$ $=$ **0 bit**

PRACTICE: COMPUTATION OF ENTROPY

- The higher the entropy, the higher the chaos



$-(1/2) * \log_2(1/2) - (1/2) * \log_2(1/2)$

$= 1$ bit

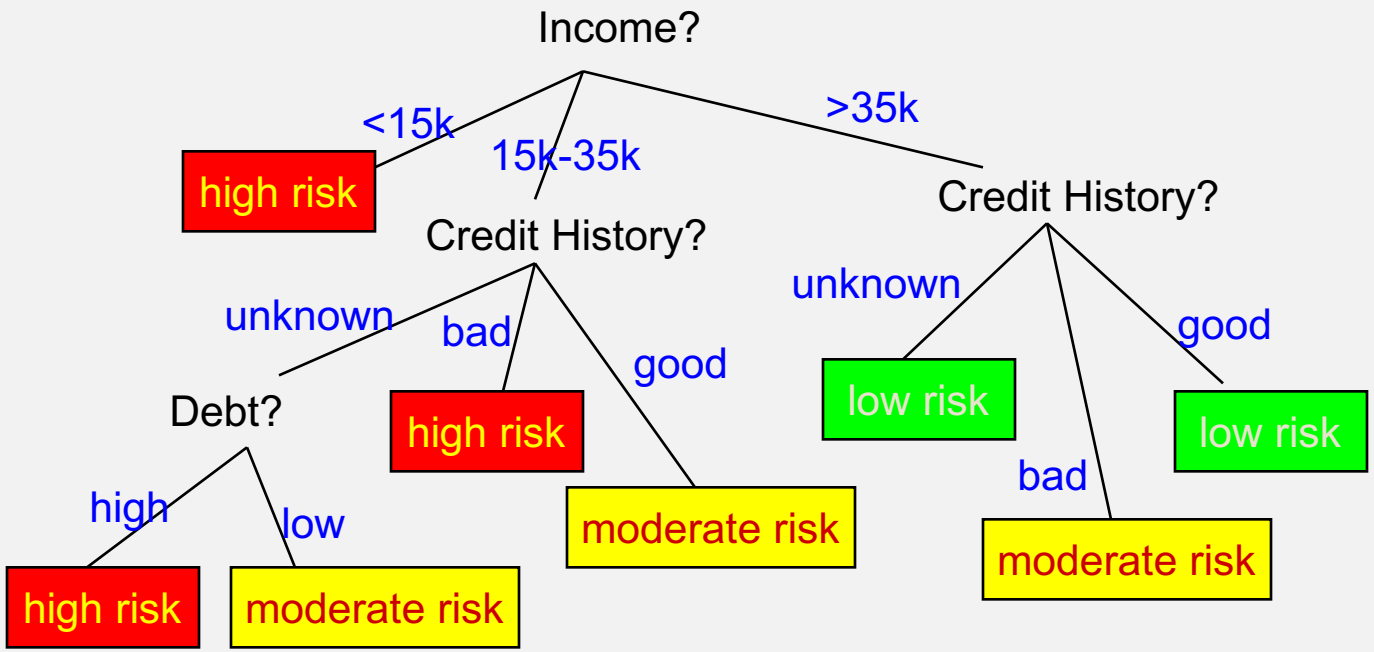
$-[(1/4) * \log_2(1/4)]*4$

$= 2$ bits

$-[(1/8) * \log_2(1/8)]*8$

$= 3$ bits

LEARNED DECISION TREE



Questions: Income? Credit History? Debt?

Classes: high risk, moderate risk, low risk

ID3 INDUCTIVE LEARNING

- Produce decision tree according to the training data
- Training set C
  - C = { E1,E2, …, E14} for root node of tree
  - P(high) = 6/14, P(moderate)=3/14, P(low)=5/14
  - Entropy for set C

$$I(C) = -(6/14) \cdot \log_2(6/14) - (3/14) \cdot \log_2(3/14)$$

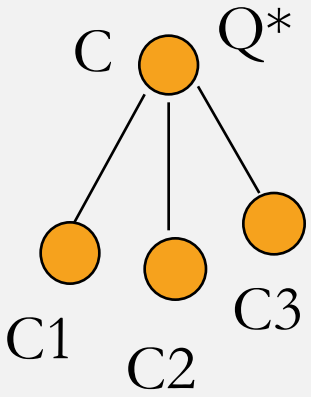
$$-(5/14) \cdot \log_2(5/14) = 1.531 \text{ bits}$$

ID3 INDUCTIVE LEARNING

- Goal of Learning Algorithm
  - The data set will be partitioned into smaller sets by applying a question.
  - In the leaf nodes, all data are of the same output category, and the partition is stopped.
    - Entropy is 0 for leaf nodes
  - Good decision tree → reduce the entropy to 0 more quickly

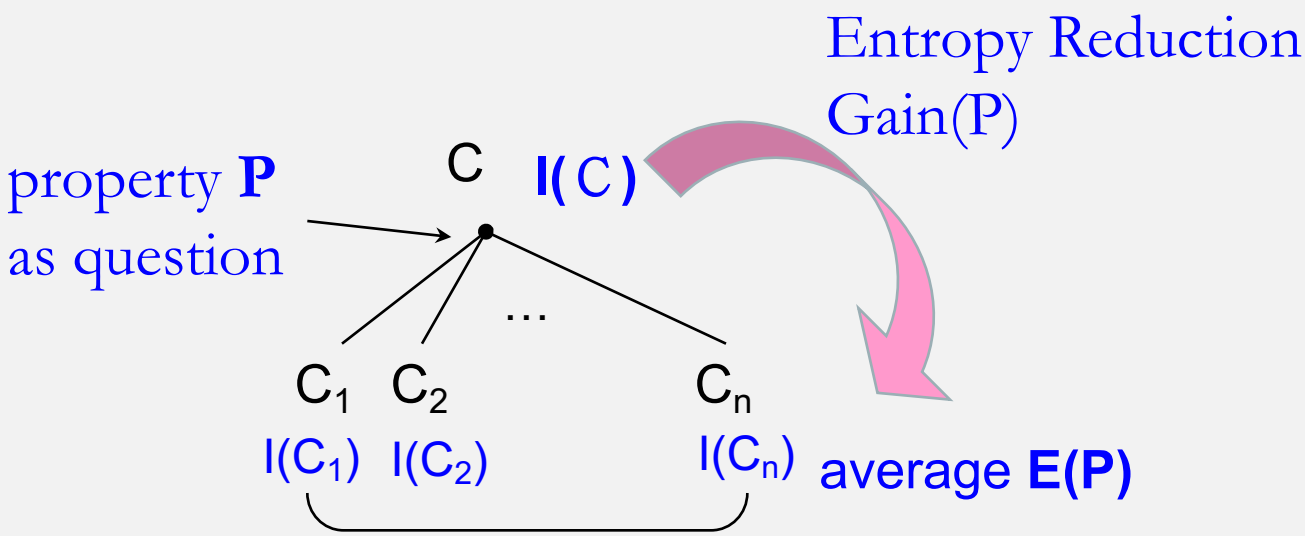
ID3 INDUCTIVE LEARNING

- For current node, a question (property) Q in a set of questions is asked (e.g. income? Collateral? Debt? 或Credit history?)
- The question Q can partition the set C into smaller sets (C → C1, C2, …, C_n)
  - E.g. income? → <15k, 15k-35k, >35k
- Which question is selected among questions?
  - Q*: maximum entropy reduction
- Once Q* is determined, apply ID3 for children (C_i's)
  - If set C_i has the entropy of 0, this set is not further processed.
  - Otherwise, call ID3 recursively for C_i with the rest of the questions.



MAXIMUM ENTROPY REDUCTION

- If S is divided into subsets $C_1 \sim C_n$ through applying the question for property P
  - $E(P) = \Sigma_1^n (|C_i|/|C|) \cdot I(C_i)$ average entropy
  - $Gain(P) = I(C) - E(P)$ entropy reduction
  - Choose P with maximum Gain for all properties



MAXIMUM ENTROPY REDUCTION

- For root node, I(C) = 1.531 bits
- Apply Q="income?" at root node, C can be partitioned
  - C1= {E1,E4,E7,E11} C2={E2,E3,E12,E14}

    C3= {E5,E6,D8,E9,E10,E13}
  - E(income) = (4/14)*I(C1)+(4/14) * I(C2)+(6/14)*I(C3)

    = (4/14)*0 +(4/14)*1 +(6/14)*0.650= 0.564 bits
  - Entropy reduction : Gain(income)=1.531-0.564 = 0.967 bits
- Similarly, other gains can be obtained
  - Gain(credit history)= 1.531-1.265 = 0.266
  - Gain(debt)=1.531-0.95 = 0.581
  - Gain(collateral) = 1.531 - 0.775 = 0.756
- Optimal question Q*="income?" (highest gain)

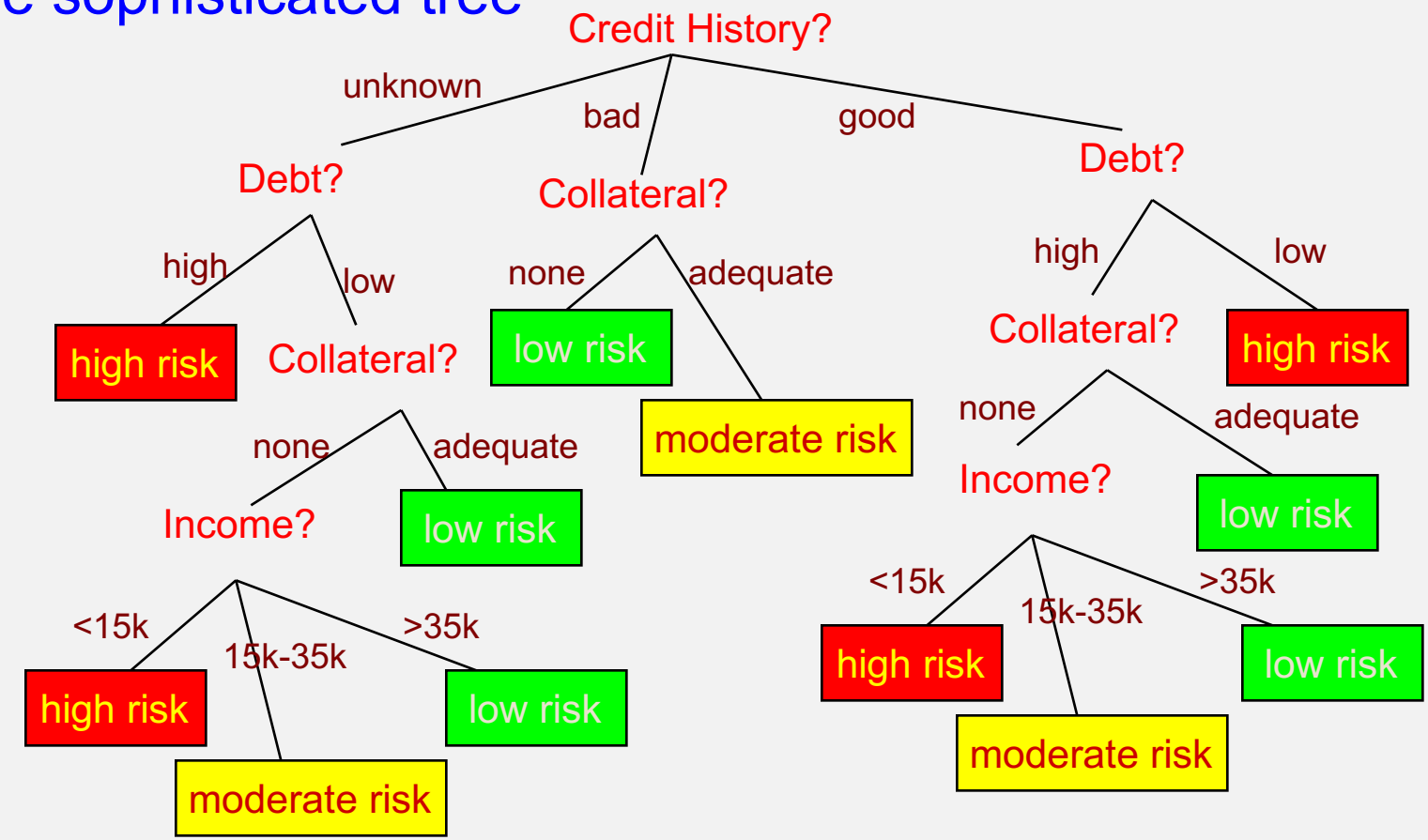
INDUCTIVE LEARNING ALGORITHM

- Training data { d_i } = { ($\underline{x}_i$, y_i) }, $\underline{x}_i$ properties, y_i class
- A set of questions Q = { P_j } based on the properties

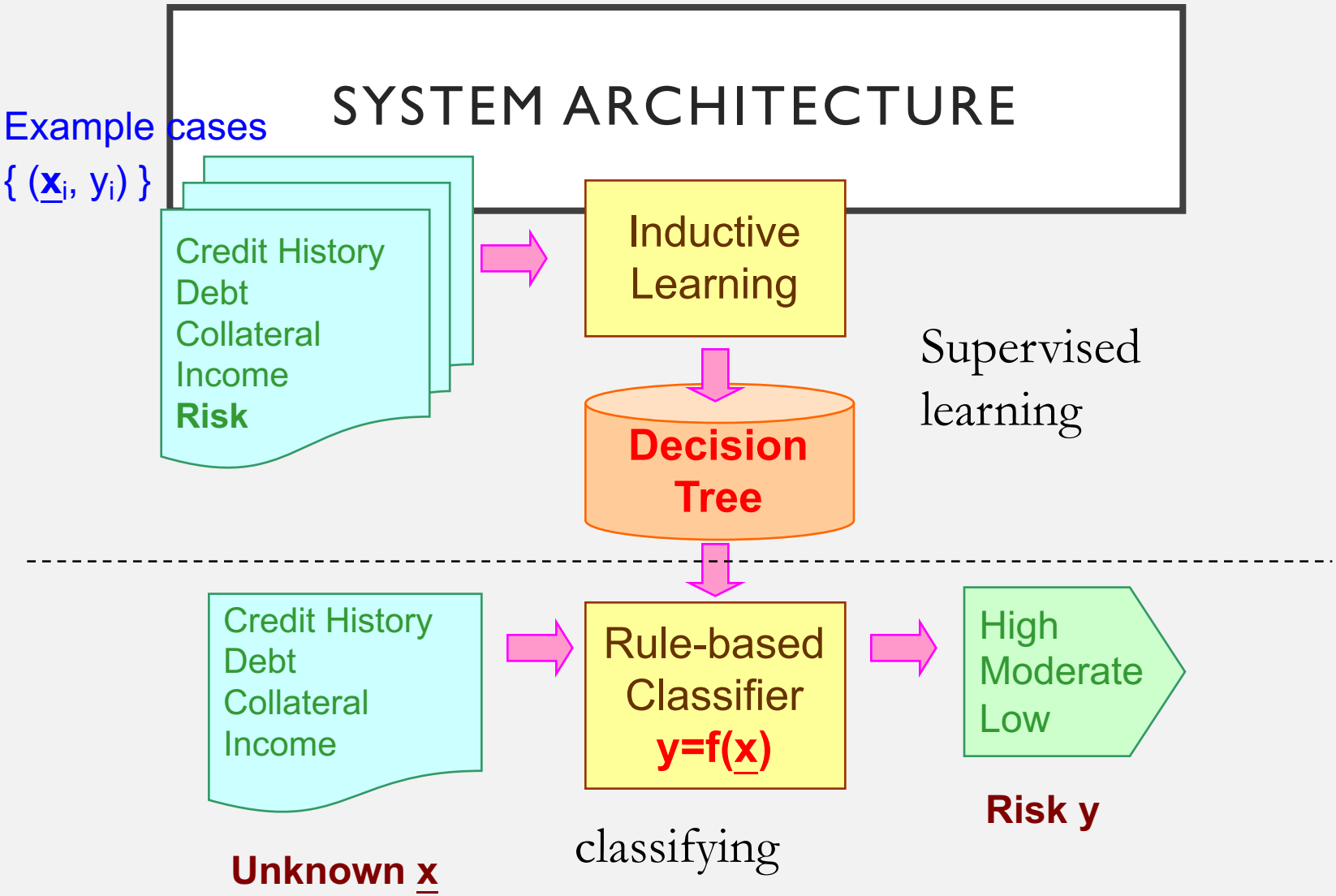
1. Initially, current set C includes all data { d_i }
2. For each d_i in C, first calculate the entropy of C, I(C). If the entropy is small, stop spanning from current set.
3. For each question P_j, generate sub-sets of C by applying question P_j. Find the entropies for these sets respectively, and calculate the average entropy $E(P_j)$.
4. Choose question P* with maximum entropy reduction as question of current set C, and apply the partition of C by P* to obtain the sub-sets.
5. Perform step 2-4 for the sub-sets respectively.

DECISION TREE IS NOT UNIQUE

More sophisticated tree



SYSTEM ARCHITECTURE



DISCUSSIONS

- A simple tree is obtained
  - Minimum depth tree
  - Greedy algorithm (select optimal question locally)
- There may be irrelevant properties
  - e.g. "Collateral" is not an relevant property and is ignored automatically because it is ineffective in entropy reduction
  - Some properties may be noises that interfere the classification if the training data is insufficient

DISCUSSIONS

- There may be classification errors
  - There might be noises (inconsistencies) for a large amount of training data (e.g. different outputs for the same input).
  - The learning is stopped if entropy is lower than a threshold (need not be 0)
  - Use majority vote to make final decision

DISCUSSIONS

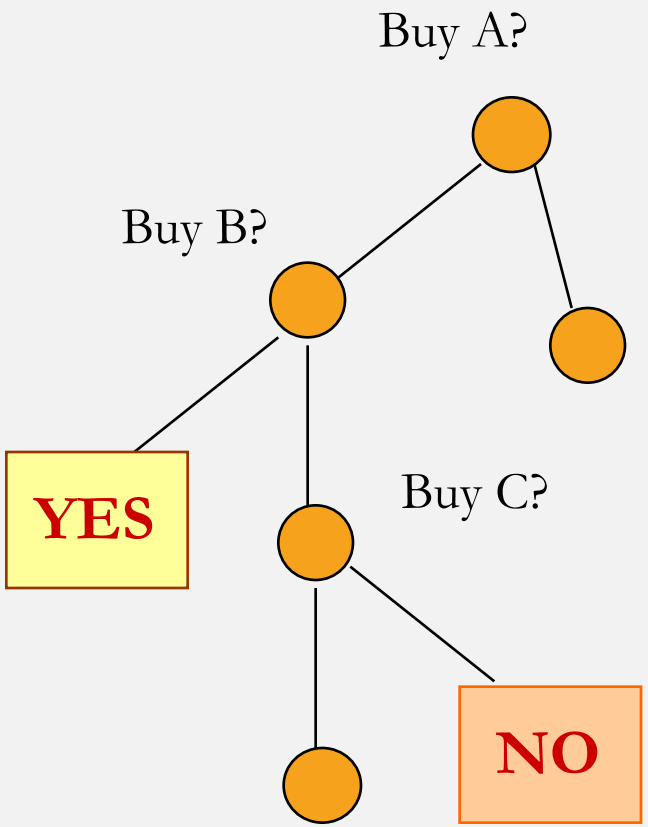
- The property need to be categorical data for ID3
  - If property P is numeric, binary test can be achieved by applying a threshold Z (such as the *income* in previous example)

    P? → P≦Z or P > Z
  - Optimal Z for property P can be obtained (maximum entropy reduction for all possible Z's)
- A question may combine several properties
  - Question: (income < 15k & debt == 'none')?
- Entropy is not the only indication of purification
  - In clustering algorithm (unsupervised learning)
  - e.g. average distance or similarity

DISCUSSIONS

- A strong bias in favor of properties with many outcomes
  - It is not fair to compare directly the properties with different number of outcomes
  - The properties with more outcomes tends to have lower average entropy because of finer partition.
- Normalization on entropy
  - Split-info(P)=$-\Sigma_i((|C_i|/|C|)\log_2(|C_i|/|C|))$
  - Gain-ratio(P) = Gain(P) / Split-info(P)
  - Maximum gain ratio instead of maximum gain
- Weka, R, Matlab

POTENTIAL COMMERCIAL APPLICATIONS



Marketing :

Who will buy product F?

→ Those who buy A & B!

Transactions

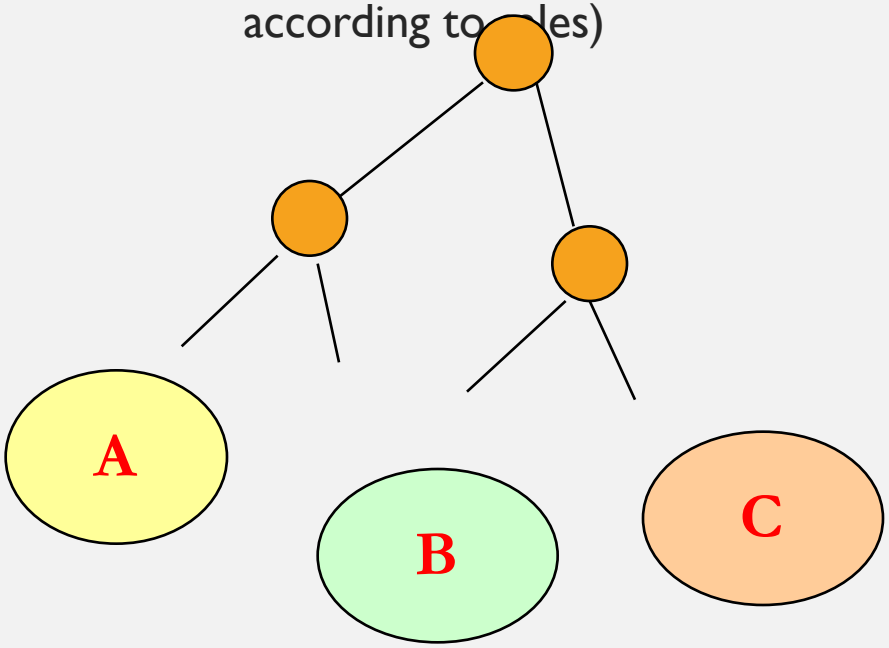
x

y

	Buy A	Buy B	…	Buy F
1	YES	YES	…	YES
2	YES	NO	…	NO
3	NO	YES	…	YES
…	…	…	…	…

POTENTIAL COMMERCIAL APPLICATIONS

- Classifying major customers with different ranks (e.g. according to sales)



APPLICATIONS

- Suitable for classifying categorical data
- If the input features contain numerical data, they need first be converted into categorical data (e.g. use regions of value)
- The knowledge learned is stored in the decision tree (or decision rules)
  - e.g. if(income > 100k && look == A) accept = true

REFERENCES

- *Data Mining: concepts, models, methods, and algorithms*

  Mehmed Kantardzic

  Wiley Inter-Science